

AN IMPROVED MODEL FOR EMOTION-AWARE COMPUTING DEVICES USING KEYSTROKE DYNAMICS AND SUPPORT VECTOR MACHINE

ENYINNAH, CHIDIEBERE

Department of Computer Science,
Babcock University, Ilishan-Remo, Ogun state, Nigeria
ebereiwobi@gmail.com

Abstract: The ability to recognize emotions is an important part of building intelligent computers. Extracting the emotional aspects of a situation could provide computers with a rich context to make appropriate decisions about how to interact with the user or adapt the system response. The problem addressed in this research is that the current methods of determining user emotion have two issues: the equipment that is required is expensive, and the majority of these sensors are invasive to the user. These problems limit the real-world applicability of existing emotion-sensing methods because the equipment costs limit the availability of the technology, and the obtrusive nature of the sensors are not realistic in typical home or office settings. This paper's solution was to determine user emotions by analyzing the rhythm of an individual's typing patterns on a standard keyboard. Keystroke dynamics approach would allow for the uninfluenced determination of emotion using technology that is in widespread use today. Field study was conducted where participants' keystrokes were collected and their emotional states were recorded via self reports. Using support vector machine technique, models based on 6 different emotional states were created. With the results from the cross-validation, best-performing emotional state models as well as other emotional states were identified that can be explored in future studies. A set of recommendations were provided for future analysis

on the existing data set as well as suggestions for future data collection and experimentation.

Keywords: Keystroke Dynamics, Emotion-aware Computing, Affective Computing, Support Vector Machine

1. INTRODUCTION

Emotion is one of the features which promotes human-computer interactions, and plays a significant role in making trust. If a recommender system can recognize the user's emotion, it would produce responses relevant to the user's emotional state. Consequently, it attracts the user's attention and loyalty. Emotion is a way of interaction which is about the message owner features. For instance, the emotional states of students at a presentation in an online tutor reflect the usability and performance of the class on students. This information even can help teachers or intelligent systems to adapt the training system for a better educational ambiance to deliver more knowledge as possible to students [11].

Emotional Intelligence can recognize human emotions and respond to the user accordingly. Human Emotions Recognition (HER) might be used in different categories such as e-learning, game, adaptive user interfaces, etc. Emotions are discussed with three parameters:

- Arousal which shows the energy of feeling.

- Valence that presents whether the feeling is a pleasure (positive) or displeasure (negative) in case of the energy.
- Dominance which shows the strength of the feelings.

Recognizing emotions of computer users are an increasingly important factor. Traditionally, computer system do not adapt according to user's emotional states. If the computer systems were capable of recognizing the emotional state that the user is going through in a particular period of time they would have many benefits for intelligent computers. Various methods have been developed and explored in order to identify emotional state. These methods are based on different parameters:

- Physiological (body temperature, heart rate)
- Visual (facial expression, gestures)
- Audio (speech, voice)
- Input devices (keystroke dynamics, mouse movement, touch screen) etc.

The importance of human emotions recognition is reflected by its wide applications in different areas such as computer games (entertainments), computer-based tutors (e-learning), machine and system controlling (industrial areas), criminology, computer privacy and security and so on [11].

Recently, the accuracy of human emotions recognition is improving by employing different available methods and techniques such as image processing, voice recognition, natural language processing, keystroke dynamics, mouse movements, touch-screen interactions, EEG devices, measuring heart rates, body heat, blood pressure and so on [13]. Some methods like image processing, voice recognition and Natural Language Processing (NLP) are time and resource consuming and they need some special circumstances such as availability of video camera

(webcam) or microphone, and some other methods require highly technical equipment such as for EEG or heart rate measuring [1]. These are medical machines which need technical training to be installed and used, and sometimes the part of skin for installation should be shaved because of the inferences and noises.

The method of emotion recognition that will be used in this work is Keystroke Dynamics. Keystroke dynamics is the study of unique timing patterns in the individual's typing and it includes extracting keystroke timing features such as the duration of key press and the time elapsed between key presses. The idea behind keystroke dynamics is that people have different typing styles and by analyzing the timings of keystrokes, a person can be identified. A benefit of this metric is that measuring keystrokes can be done through a keyboard, thus negating the cost of typical physiological biometric systems which require expensive hardware to measure physical attributes.

There have been various methods for evaluating emotional states like facial expression analysis, voice intonation analysis, change in breathing and physiological sensors attached to the skin etc. that have varying rates of success, but they still exhibit one or both of two main problems preventing wide scale use: They can

- Be intrusive to the user
 - Require specialized equipment that is expensive and not found in typical home or office environments [6].
- Compared to other biometric modalities, keystroke biometrics has more desirable properties due to being user-friendly and non-intrusive.

Keyboard keystroke dynamics, mouse movement and touch-screen interaction are the three methods that are mostly available on every computer or even portable digital devices such as handhelds, tablets, and mobile

phones. This work's focus is on the use of keyboard for human emotions recognition because of availability and popularity of the equipment.

There are three problems which are usually ignored in most of the researches for human emotion recognition in computer systems:

- Human emotions are not constant and they do not occur and exist in one single level. For instance, if you feel happiness for two days, the level and strength of feeling would not be the same for every moment that you have the same emotion. It means that emotions are more than being represented in 0 and 1. Nowadays, most of research papers present detection methods of emotion whether they exist or not. For example, if the user is happy or not; but it does not define the level of possible happiness [19]. Emotions have a fuzzy basis, and a discrete calculation would not be a solution on it.
- Users as human beings may not have only one emotion at a time; they may have different emotions in different levels. This second fact is also ignored in the most of researches, and the current human emotions recognition systems attempt to find only one current emotion and perhaps the strongest emotion at that time [20]. It is very clear that many emotions influence our life style and we are living with the combination of emotions. The happiness at the time of receiving a gift and the happiness at the time being in a theme park can be at the same level, but they are presented in two different qualities because the other available emotions are different in those two cases [6].
- Psychological studies show differences in human emotional changes pattern based on cultural and language backgrounds. These differences say that a specific emotional pattern which is extracted from a specific area cannot be extended to the other part of the world Lewis.

This work analyzes human emotions based on a Support Vector Machine which is able to recognize the human emotions in six levels:

- Anger
- Fear
- Disgust
- Happiness
- Sadness
- Surprise

2. LITERATURE REVIEW

The terms affect, mood, and emotion can be confusing and are often used interchangeably; however, it is important to understand the distinction between these terms. In this paper, affect is used in a more general sense that encompasses both mood and emotions [8]. Moods are subtle, long in duration, and are usually spoken of in general terms. The subtle nature of moods can mean that they go unnoticed to the person experiencing them until their mood is brought to their attention. In contrast, emotions are usually reactionary; they are often triggered by some particular cause either physical or cognitive and are short in duration. Also, the individual is usually aware of the presence of an emotion [21].

There are two different approaches that have traditionally been taken in describing emotion: those that emphasize the cognitive (mental) aspects and those that emphasize the bodily (physical) aspects. The cognitive approach has been attributed to Walter Cannon who suggested that emotion is experienced within the brain, independently of the sensations of the body [4]. The physical approach focuses on the physiological response (e.g. elevated heart rate) that occurs just prior or during an emotional episode; this approach has largely been attributed to William James [21].

Recent approaches see emotion as the combination of these two aspects (cognitive and physiological) simultaneously contributing to emotion. Thoughts as well as changes in body chemistry alone can cause emotions to arise in individuals [21]. For example, [22] suggests that emotion is the result of our interpretation of our bodily responses and our situation, which we attach a label to (e.g. fear). When emotional state is referred to in this paper, it means the internal dynamics (both cognitive and physiological) that are present during an emotional episode and we describe the emotional experience as what an individual perceives of their emotional state [21].

Emotional Experimentation Environment

There are two main approaches to collect emotion data. One approach is to induce moods in the participants in a laboratory setting and then collecting the desired data. This requires a mood induction procedure (MIP). A number of MIP categories exist [30], including imagination, social interaction and using film or story. While MIPs have the advantage of being able to control the moods of the participant, the experiments often take a considerably long time and do not always guarantee successful induction. Furthermore, individuals may react differently in a laboratory setting compared to a more naturalistic setting because the experimental situation may influence the individual [18]. A naturalistic setting during an experiment aims to observe participants in their natural setting to have minimal influence on the participant's behavior. The experience sampling methodology (ESM) [9] is such an approach. In this technique a participant's experiences are recorded as they occur at certain moments in time. This has the advantage that it can capture daily life from moment to moment without the problem of recall issues and that it is much easier for participants to indicate their experiences at the

moment that they are experiencing them. A main disadvantage of using a naturalistic setting is that all techniques provide the experimenter with subjective information. This could lead to biased information and individuals may repress certain information or change their responses to fit the norm of the participant's culture.

Keystroke Dynamics

Keystroke dynamics is the study of the characteristics that are present in a user's typing rhythm when using a keyboard. An individual's typing behavior can change when the individual experiences different emotions. This means that information about a user's typing behavior may allow us to infer the user's emotional state. Two main approaches for keystroke dynamics analysis exist: fixed text analysis and free text analysis. The former usually requires participants to type one or more fixed pieces of text multiple times during the data collection process. Using fixed text to build a model implies that this model can only be used at those moments that the user types one or more of the fixed pieces of text on which the model was trained. The latter allows any sequence of text as input. This enables the model to be used during continuous monitoring, which is very desirable for affecting recognition because this means that emotional information is available in the computer system at any time. Furthermore, using this approach it is also possible to obtain the keystroke data unobtrusively, which is also highly desirable. Typical keystroke dynamics features are keypress duration and latencies between multiple keystrokes.

Text Content Analysis

The actual content of text contains a lot of valuable information concerning affective state. A possible starting point of a linguistic analysis of text to extract emotional

information is to use specific affective lexicons. Examples of such lexicons are the Affective Norms for English Words (ANEW), SentiWordNet and WordNet Affective state. [5] argued that words need to be distinguished based on the fact whether they directly refer to emotional states or contain an indirect reference that depends on the context. A more abstract analysis using specific textual features is also possible. [29] used a number of features defined by [33]. It is also possible to calculate features based on annotated datasets and using these in a machine learning algorithm.

Mouse Behavior

Except for text content and keystroke dynamics, also mouse behavior may contain useful information on an individual's emotional state. One can distinguish single clicks, double clicks, either using the left, right or maybe middle button. Furthermore, one can observe mouse movements and mouse wheel movements. From the mouse movement data, the distance, angle and speed features between pairs of data points can be extracted.

Emotion Recognition

There have been numerous studies investigating the possible applications of keyboard and mouse behavior in authentication and security but for this work the applications regarding emotion recognition are more interesting. [32] described a method to correlate keyboard and mouse interaction with affective states using a MIP and the PA emotion model. Later, [29] proposed a new way of assessing cognitive and physical stress by analyzing keystroke dynamics using both content and timing features. They achieved correct classification rates of 62.5% for physical stress and 75% for cognitive stress. [6] focused more on actual emotions and investigated the possibility to identify different emotional states using keystroke analysis. They focused on gathering keystroke

data in a natural context using the experience sampling methodology rather than a laboratory environment. They achieved reliable accuracy rates for fixed text ranging from 77.4% to 87.8% for confidence, hesitance, nervousness, relaxation, sadness and tiredness. [25] also introduced a method to detect student boredom during the attendance in an online lesson that was followed in a laboratory setting and achieved a correct classification rate for fixed text above 90%. Except for analyzing keystroke and mouse features, it is also possible to use other additional sensors that are present in today's smartphones. [16] performed an experimental study, which they extended in, to classify different moods using this kind of data. Continuing in the area of smartphones, Lee et al. investigated the possibilities to automatically recognize emotions in social network service posts. They achieved an average classification accuracy of 67.5%. Later, [26] validated the hypothesis about the existence of the difference in typing patterns between different emotional states using the facial feedback hypothesis. [20]. focused on textual contents as well as keystroke dynamics. They used WordNet and the ISEAR dataset in combination with a vector space model. [10] presented a method to evaluate a user's boredom and frustration in an intelligent learning environment but focused on free text in contrast to. Recently, [17] conducted another study that examined the variance in keystroke typing patterns caused by emotions using visual stimuli to induce emotional states. They concluded that the effect of emotion is significant but small compared to the individual variability. Along with the increasing popularity of fuzzy logic, [23] and [2014] experimented with fuzzy models. Finally, a number of meta-studies were conducted by [13] and also a number of interesting possible applications were presented, their initial research. Gaines et al. focused

mainly on professional typists, that is, secretaries with formal training. Authentication studies since then have included participants of variable typing skills. Their research suggests that even with the very fast typing speeds of expert typists (very small timing values); participants were still able to be successfully identified by their keystroke dynamics. Others have found that the most inconsistent classifications come from models that were built for novice typists. Novice users, such as those that use only the two fore-fingers while typing seemed to have very inconsistent typing patterns with long pauses in between keystrokes.

Some studies included an outlier removal process to account for differences in participants' typing abilities as well as other naturally-occurring behaviors such as long pauses or breaks in typing. Early studies such used a single low-pass filter to remove these outliers from the data. Unfortunately, one single filter does not perform very well across all participants due to differences in typing abilities. For instance, when looking at the key duration of an expert typist, the timing values (duration and latency) are likely to be shorter than those of a novice user. Outliers in the expert typist's timing values could resemble typical novice user values. Depending on the threshold used to remove outliers, the threshold will be either too specific (removing the valid novice users' data) or too general (with no outliers removed for the expert user).

To avoid this problem, some studies used separate threshold values for each participant to accommodate for each user's typing proficiency. One common technique that was used was to calculate the mean and standard deviation from an individual's typing samples, and then any data that was three standard deviations away was

removed from the data set and the mean and standard deviations were then recalculated on the remaining data . [27] illustrated the importance of threshold removal by plotting different threshold values against the classification rates to determine the optimum threshold to use during their outlier removal process. Other studies provided different results by segmenting portions of their populations to handle these differences; for example, in [27] segmentations were performed based on whether the participant was using a laptop or desktop. However, as Ekman suggests, these types of scenarios may be unrealistic in real-world authentication systems. They claim that by fine-tuning the model to the data, the authors over-fit the model to the training data set.

The keystroke dynamics research presented above consisted of authentication and verification systems as this was the area where the majority of the research originated from. In this section, we look at the studies that combine affective computing and keystroke dynamics.

[2003] described a methodology that could be used to find correlations between user interactions (keyboard and mouse) and their affective state. In this paper, he described an empirical study where he used film clips to induce participants into various states along the arousal and valence dimensions. To determine the affective state of the participants, physiological sensors were used to measure respiration, pulse, skin conductance level, and corrugator (the small muscle that controls part of the eyebrow) activity. Participants were also asked, at different times, to self-assess their current emotional state using the Self-Assessment Manikin (SAM), devised by Lang. The authors found significant differences between the neutral state when compared to the other emotional states; however, they were unable to distinguish between the other four states that were induced.

A study performed by Hancock et al., looked at emotional contagion between participants in computer-mediated communication. In this empirical study, three induction techniques were used to induce a negative (sadness and frustration) affect state: a video clip, music, and an additional task that was either easy or hard depending on the condition being tested. The emotional state of the participants was assessed using a questionnaire with 7-point Likert scale statements that the users were asked to complete. The authors created profiles based on linguistic patterns produced by the Linguistic Inquiry and Word Count (LIWC) software package which looks at word frequency along different psychological dimensions. They found that individuals could identify the negative affective state in their partners through text-based cues and that emotional contagion also took place between the participants in the negative affect condition.

Recent work in affective computing using keystroke dynamics has been performed by Visor et al. where keystroke and linguistic features were used to identify both cognitive and physical stress. In this empirical study, free text was collected and five different data mining techniques were built using features that focused on the words used in the sample, timing and keystroke features. They achieved classification rates of 62.5% for physical stress and 75% for cognitive stress, which they state was comparable to other proposed affective computing solutions. The authors mentioned that although their results indicate that they can detect stress through keystroke dynamics using their empirical methodology, the effectiveness should also be tested across a wider range of typing abilities, cognitive/physical abilities, and keyboard types, as well as in real-world stressful situations.

This study attempts to address the real-world (ecological validity) aspect that was missing in the related literature.

3. THE PROPOSED SOLUTION

The proposed methodology was to develop an application using the prototyping system development process model to develop an application that will be used to capture keystroke data from selected users and log such data for feature extraction and analysis. The idea was to create an environment in which the keystrokes of selected users are collected for analysis with minimum interference to facilitate for proper analysis.

The collected keystrokes were then analyzed using the support vector machine algorithm to determine emotional states of users from the captured keystrokes. The support vector machine algorithm was used with the intention of identifying the following emotional states:

- Anger
- Fear
- Disgust
- Happiness
- Sadness
- Surprise

As these are very common emotions individuals exhibit. It must be pointed out here that this research work, like others conducted in this problem domain, focused on detecting states of emotion rather than levels of a particular emotion throughout a particular period. It is generally agreed that dealing with levels of a particular emotion in a period is difficult to analyze as the human emotion is neither stable nor constant. A simple example of such emotional levels is level of happiness over say a period of twenty four hours (24hr).

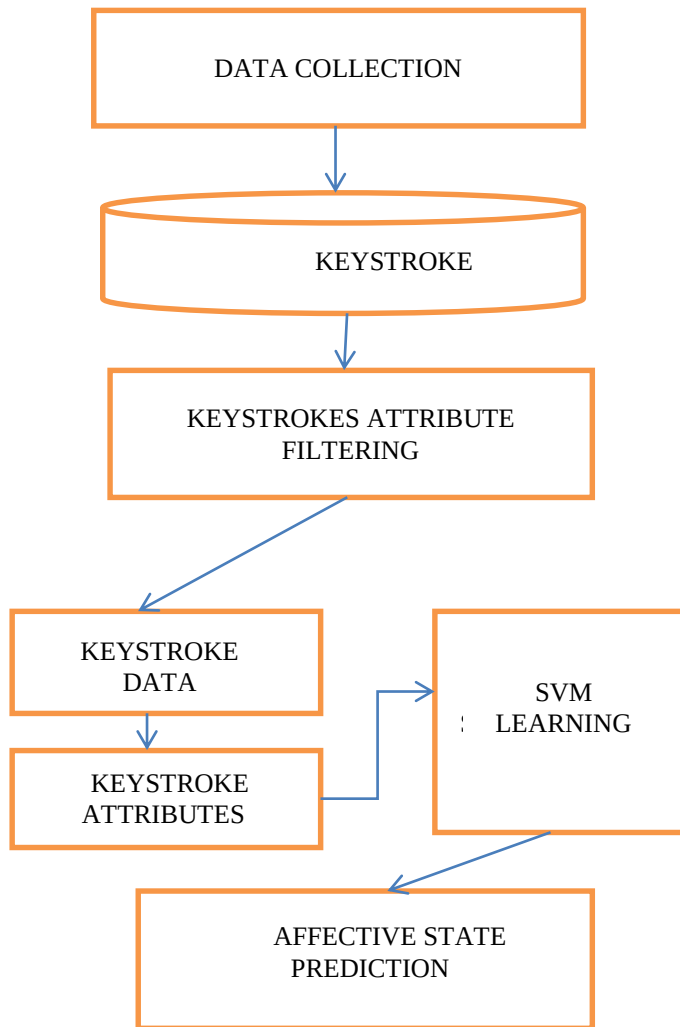


Figure 1: The proposed methodology

The idea that you or anyone could be happy at a particular level of emotion for the whole 24hr would be false as the level of happiness would most certainly be different at any particular point in time within that 24hr period. The same thing goes for the other stated emotions hence the need to carry out this research based on states of emotions rather than levels of emotion.

The advantage to identifying affective states using keystroke dynamics is that it avoids some of the previously-described issues found in affective state determination research such as the expense, intrusion and

use of specialized hardware. Keystroke dynamics for the investigation of emotional states rather levels of emotion are more effective both in terms of cost and accuracy. Keystroke logging is very unobtrusive to the user and is undetectable by the average user without the aid of special computer programs. This is advantageous when measuring emotional state as it should reduce the interference effect of the data collection on the user's true emotional state.

To achieve the above discussed methodology, the proposed solution followed the steps below:

Sample Space

The purpose of this methodology was to gather temporal feelings in the moment 'rather than retrospectively, and to gather real world data instead of inducing emotional states in a lab. The data collection software was installed on participants' computers for a period of time. The participants were free to use their computer as they normally would so as to gather accurate emotional data rather than induced, but were asked to fill out an emotional state questionnaire when the software prompted them. The emotional state questionnaire contained questions on the above stated emotional states.

Data Collection Software

Software was developed to collect keystroke data by the user or participant of the experiment. The software is designed in such a way as to be easy to use and not obstructive to the user. Based on the activity of the user, the participants were asked by the software to fill out a short self-report on their current emotional state. The keystroke data as well as the answers to the self-report were then collected and stored for further processing

Post Processing and Feature Extraction

Feature	Meaning
Num_char	Number of characters pressed in every 5 sec
mode_dwell	Mode of dwell times of keys pressed in every 5 sec
stdDev_dwell	Standard deviation of dwell times of keys pressed in every 5 sec
stdVar_dwell	Standard variance of dwell times of keys pressed in every 5 sec
range_dwell	Range of dwell times of keys pressed in every 5 sec
min_dwell	Minimum of dwell times of keys pressed in every 5 sec
max_dwell	Maximum of dwell times of keys pressed in every 5 sec
mode_flight	Mode of flight times of keys pressed in every 5 sec
stdDev_flight	Standard deviation of flight times of keys pressed in every 5 sec
stdVar_flight	Standard variance of flight times of keys pressed in every 5 sec
range_flight	Range of flight times of keys pressed in every 5 sec
min_flight	Minimum of flight times of keys pressed in every 5 sec
max_flight	Maximum of flight times of keys pressed in every 5 sec
mode_d2d	Mode of down-to-down times of keys pressed in every 5 sec
stdDev_d2d	Standard deviation of down-to-down times of keys pressed in every 5 sec
stdVar_d2d	Standard variance of down-to-down times of keys pressed in every 5 sec
range_d2d	Range of down-to-down times of keys pressed in every 5 sec
min_d2d	Minimum of down-to-down times of keys pressed in every 5 sec
max_d2d	Maximum of down-to-down times of keys pressed in every 5 sec

Upon completion of the data capture using the developed software, the data was collected from the key logger. Data cleaning was performed and identified features were extracted from the raw keystroke data. The features were then extracted using the necessary algorithm for processing the key log. Due to the large number of features that was extracted, attribute filtering was needed to select some certain attributes to facilitate machine learning processing.

A. EXTRACTING KEYSTROKE FEATURES

Extracting keystrokes was very important to the model. The extracted keystroke is based on the logged keystroke data that has been captured by the selected

software. Table 1 shows a sample keystroke extraction feature.

Table-1: Extracted Keystroke features and their meaning.

The table shows keystrokes attributes and their meaning, this was built into the software for extraction and analysis purposes. It must be stated here that not all keystrokes are actually useful hence the need to select attributes.

Keystroke dynamics introduces an additional parameter to the affective state detection process: something that qualifies the user or his behavior (i.e., the way of typing phrases). Using this additional parameter strengthens the affective state detection process. The capture process is presented in Figure 2. It consists in capturing several features when the keys are pressed and released (timestamp of the event, code of the key, . . .).

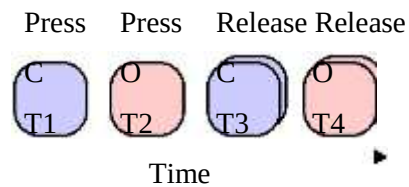


Figure 2: Information capture in a keystroke dynamics system when pressing C and O keys

The features extraction consists mainly in computing different latencies and duration times between each key. Figure 2 shows an example where the user presses two keys of the keyboard. The user presses "C" at T1, "O" at T2 and releases "C" at T3 and "O" at T4. Note that the following relation is always respected: $T_3 > T_1$ and $T_4 > T_2$ (always release a key after pressing it), while the following condition may not always be respected: $T_2 >$

T 3 (because, as in the study example, a user may press another key before releasing the previous one). The study can extract three different types of latencies (T2-T1, T4-T3, T2-T3) which will be called PP (latency between two pressures), RR (latency between two releases), RP (latency between one release and one pressure) respectively and one type of duration (T3-T1 or T4-T2) which will be called PR (duration of a key press). In the example, T2-T3 is negative because the user presses O before releasing C (this happens frequently when a user types fast). This is not always true, but it is quite discriminating. The described process is repeated for all the keys.

This process of capturing keystrokes is adopted for the research work because it has been proven to yield the best result in terms of detecting affective states of the user at the time of time a phrase.

B. SELECTED ATTRIBUTES

This involves searching through a possible or all possible set of attributes so as to find a subset of attributes that works best for prediction purposes. To do this, two objects must be set up: an attribute evaluator and a search method. The evaluator determines what method is used to assign a worth to each subset of attributes. The search method determines what style of search is performed. These attributes will be selected based on some set of parameters which includes the following:

1. Typing speed in 5 second interval,
2. Minimum, maximum, standard variance, mode and range of dwell time of each character pressed in every 5 seconds.

3. Minimum, maximum, standard variance, mode and range of flight time of each character in 5 seconds interval.

C. SUPPORT VECTOR MACHINE CLASSIFICATION

Support vector machine are supervised learning models with associated learning algorithms that analyze data used for classification and regression analysis. Given a set of training examples, each marked as belonging to one or the other of two categories, an SVM training algorithm builds a model that assigns new examples to one category or the other, making it a non-probabilistic binary linear classifier. An SVM model is a representation of the examples as points in space, mapped so that the examples of the separate categories are divided by a clear gap that is as wide as possible. New examples are then mapped into that same space and predicted to belong to a category based on which side of the gap they fall.

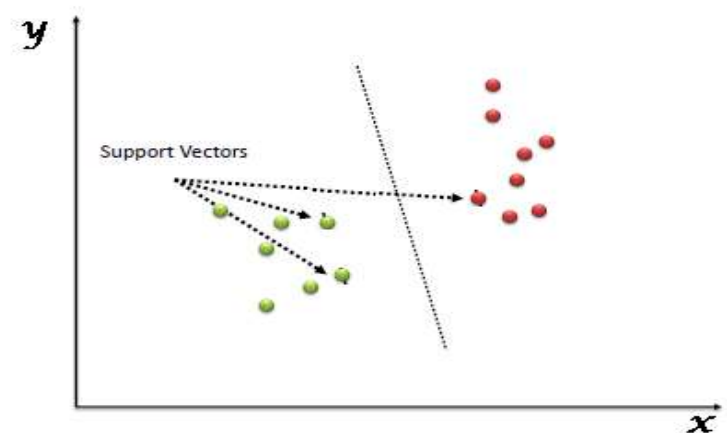


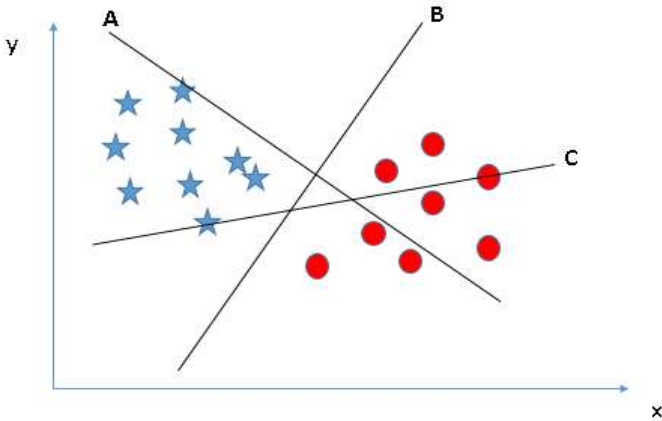
Figure 3: the segregation of two classes with a hyper-plane

In addition to performing linear classification, SVMs can efficiently perform a non-linear classification using what is called the kernel trick, implicitly mapping their inputs into high-dimensional feature spaces.

When data are not labeled, supervised learning is not possible, and an unsupervised learning approach is required, which attempts to find natural clustering of the data to groups, and then map new data to these formed groups. Figure1 shows the segregation of two classes with a hyper plane.

Each emotion class contains several vectors. To build a model of a specific emotion class, the mean of all these vectors will be taken. The final vector is the model for that emotion class. In this way all 5 models were built for 5 emotional classes. Finally a $5 \times n$ matrix is built which I will denote as matrix M . to get an idea of this vector classification, let's consider the following:

Let's say we have a set of star and circle character objects to be classified and let's also assume we have three (3)



vectors of hyper-planes(A,B, and C), then the problem would be how to choose the best support vector out of the three planes.

Figure 4: hyper-planes A, B and C

To do this, we need to remember a thumb rule to identify the right hyper-plane which is: “Select the hyper-plane which segregates the two classes better”. In this scenario, hyper-plane “B” has excellently performed this job.

To test a query text, the query document is also converted to a vector containing all term weights. This is then converted to an $n \times 1$ matrix which we denote as Q . Q is multiplied with M row by row and 5 numeric values are obtained. The highest value of these numbers is the index of a particular emotion class in and is the indicator of that class. So the query text belongs to that emotion class. So what is being done in essence is selecting the one that properly represents the class, just like the vector hyper-plane.

How is the optimal hyper-plane computed?

Let's introduce the notation used to define formally a hyper-plane:

$$f(\mathbf{x}) = \beta_0 + \beta^T \mathbf{x},$$

Where β is known as the *weight vector* and β_0 as the *bias*.

The optimal hyper-plane can be represented in an infinite number of different ways by scaling of β and β_0 . As a matter of convention, among all the possible representations of the hyper-plane, the one chosen is

$$|\beta_0 + \beta^T \mathbf{x}| = 1$$

Where $\mathbf{x}$ symbolizes the training examples closest to the hyper-plane. In general, the training examples that are closest to the hyper-plane are called **support vectors**. This representation is known as the **canonical hyper-plane**.

Now, we use the result of geometry that gives the distance between a point $\mathbf{x}$ and a hyper-plane (β, β_0) :

$$\text{distance} = \frac{|\beta_0 + \beta^T \mathbf{x}|}{\|\beta\|}.$$

In particular, for the canonical hyper-plane, the numerator is equal to one and the distance to the support vectors is

$$\text{distance}_{\text{support vectors}} = \frac{|\beta_0 + \beta^T x|}{\|\beta\|} = \frac{1}{\|\beta\|}.$$

Recall that the margin introduced in the previous section, here denoted as M , is twice the distance to the closest examples:

$$M = \frac{2}{\|\beta\|}$$

Finally, the problem of maximizing M is equivalent to the problem of minimizing a function $L(\beta)$ subject to some constraints. The constraints model the requirement for the hyper-plane to classify correctly all the training examples x_i . Formally,

$$\min_{\beta, \beta_0} L(\beta) = \frac{1}{2} \|\beta\|^2 \text{ subject to } y_i(\beta^T x_i + \beta_0) \geq 1$$

Where y_i represents each of the labels of the training examples.

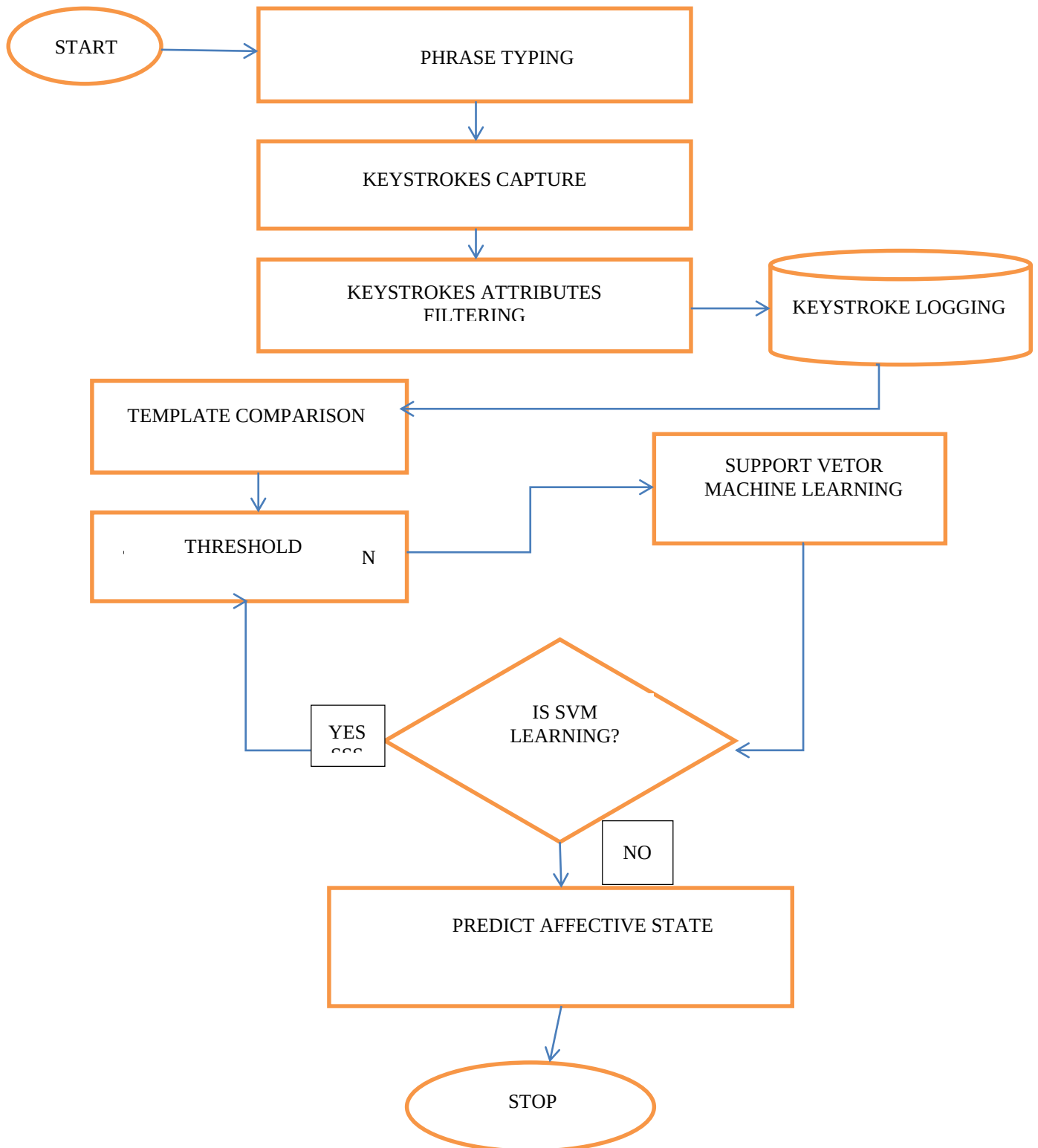
This is a problem of Lagrangian optimization that can be solved using Lagrange multipliers to obtain the weight vector β and the bias β_0 of the optimal hyperplane.

Figure 5 shows a sample algorithm in OpenCV for the computation of the optimal hyper-plane using the above discussed procedure.

```
int main()
{
    // Data for visual representation
    int width = 512, height = 512;
    Mat image = Mat::zeros(height, width, CV_8UC3);
    // Set up training data
    float labels[4] = {1.0, -1.0, -1.0, -1.0};
    Mat labelsMat(4, 1, CV_32FC1, labels);
    float trainingData[4][2] = { {501, 10}, {255, 10}, {501, 255}, {10, 501} };
    Mat trainingDataMat(4, 2, CV_32FC1, trainingData);
    // Set up SVM's parameters
    CvSVMParams params;
    params.svm_type = CvSVM::C_SVC;
    params.kernel_type = CvSVM::LINEAR;
    params.term_crit = cvTermCriteria(CV_TERMCRIT_ITER, 100, 1e-6);
    // Train the SVM
    CvSVM SVM;
    SVM.train(trainingDataMat, labelsMat, Mat(), Mat(), params);
    Vec3b green(0,255,0), blue(255,0,0);
    // Show the decision regions given by the SVM
```

```
for (inti = 0; i < image.rows; ++i)
for (int j = 0; j < image.cols; ++j)
{
    Mat sampleMat = (Mat_<float>(1,2) << j, i);
    float response = SVM.predict(sampleMat);
    if (response == 1)
        image.at<Vec3b>(i,j) = green;
    elseif (response == -1)
        image.at<Vec3b>(i,j) = blue;
}
// Show the training data
int thickness = -1;
int lineType = 8;
circle( image, Point(501, 10), 5, Scalar( 0, 0, 0), thickness, lineType);
circle( image, Point(255, 10), 5, Scalar(255, 255, 255), thickness, lineType);
circle( image, Point(501, 255), 5, Scalar(255, 255, 255), thickness, lineType);
circle( image, Point( 10, 501), 5, Scalar(255, 255, 255), thickness, lineType);
// Show support vectors
thickness = 2;
lineType = 8;
int c = SVM.get_support_vector_count();
for (inti = 0; i < c; ++i)
{
    const float* v = SVM.get_support_vector(i);
    circle( image, Point( (int) v[0], (int) v[1]), 6, Scalar(128, 128, 128), thickness, lineType);
}
imwrite("result.png", image); // save the image
imshow("SVM Simple Example", image); // show it to the user
waitKey(0);
}
```

Figure 5 shows the system flowchart of the proposed solution. Data needed to analyze keystroke dynamics is obtained by keystroke logging. Normally, all that is retained when logging a typing session is the sequence of characters corresponding to the order in which keys were pressed and timing information is discarded.

Figure 5 System Flowchart

4. EXPERIMENTAL ANALYSIS

This section of the research work discusses the details of the experiments carried out. This study will be comparing its results with that of reputable researchers in the field to show that the proposed model performs well when compared with other related works. The study will also design a simulator to run the proposed algorithm. The simulator is designed to take advantages of the leveraging parts of the proposed model and also to give a visual display of the algorithm at work.

A. THE SIMULATOR

The simulator was designed using the Microsoft C# programming language and the Microsoft visual studio integrated development environment. The simulator is comprised of different modules which include:

1. The keystrokes capture module,
2. The Load data module,
3. Support Vector Machine analysis module,
4. The data display module,
5. The chart module,
6. The result modules which comprises:
Emotion determination and Confusion matrix.

The simulator utilizes the above mentioned modules to achieve the goal of the research work. A brief discussion of the modules is given below while a more detailed discussion is given in the sections that follow:

- **Keystroke Capture Module:** This module is used to capture keystroke dynamics as the user of

the application keys in data displayed by the application.

- **Load Data Module:** This module is used to load the captured keystroke into a Microsoft SQL Server database server for processing.
- **Support Vector Machine Analysis Module:** This module processes the uploaded data using the support vector machine algorithm.
- **Display Data Module:** This module displays the uploaded keystrokes dynamics that was uploaded in the load data module.
- **Chart Module:** This module shows a chart comparison of different affective state identification for the processed keystroke dynamics data.
- **Result Module:** This module is comprised of two module, emotion identification and the Confusion matrix.

B. SIMULATOR MODULES DISCUSSION

This section discusses the simulator modules in detail.



Figure 6: Keystroke Dynamics Capture Module

Figure 7 shows the interface of the module. The interface comprises of components to capture keystrokes, this components include a read only rich text box which displays the data to be keyed in by the user of the application.

Figure 6 Simulator interface for capturing and exporting keystroke dynamics

Figure 6 also contains another rich text box that the user types into and a group of select boxes that is used to capture the affective state of the user while typing the displayed data. The user clicks the “done” button and each keydown and keyup time interval for each button click is logged in a note pad. The features extraction consists mainly in computing different latencies and duration times between each key. Drawing from section 3.3 of this research work, it can be seen how this difference in latency of keystroke can be extracted. Say a user (from the example of section 3.3) presses “C” at T1, “O” at T2 and releases “C” at T3 and “O” at T4. Note that the following relation is always respected: $T_3 > T_1$ and $T_4 > T_2$ (always release a key after pressing it), while the following condition may not always be respected: $T_2 > T_3$ (because, as in the study example, a user may press another key before releasing the previous one, like when a user presses control “ctrl” key and “A” key to select all text. i.e. “ctrl + A”). Three different types of latencies can be extracted ($T_2 - T_1$, $T_4 - T_3$, $T_2 - T_3$) which will be called PP (latency between two pressures), RR (latency between two releases), RP (latency between one release and one pressure) respectively and one type of duration ($T_3 - T_1$ or $T_4 - T_2$) which will be called PR (duration of a key press). In the example, $T_2 - T_3$ is negative because the user presses O before releasing C (this happens frequently when a user types fast). This is

Feature	Meaning
Num_char	Number of characters pressed in every 5 sec
mode_dwell	Mode of dwell times of keys pressed in every 5 sec
range_dwell	Range of dwell times of keys pressed in every 5 sec
min_dwell	Minimum of dwell times of keys pressed in every 5 sec
max_dwell	Maximum of dwell times of keys pressed in every 5 sec
mode_flight	Mode of flight times of keys pressed in every 5 sec
range_flight	Range of flight times of keys pressed in every 5 sec
min_flight	Minimum of flight times of keys pressed in every 5 sec
max_flight	Maximum of flight times of keys pressed in every 5 sec
mode_d2d	Mode of down-to-down times of keys pressed in every 5 sec
range_d2d	Range of down-to-down times of keys pressed in every 5 sec
min_d2d	Minimum of down-to-down times of keys pressed in every 5 sec
max_d2d	Maximum of down-to-down times of keys pressed in every 5 sec

not always true, but it is quite discriminating. The described process is repeated for all the keys.

This above process is used by the keystroke dynamics capture module to capture and log these differential latencies in a note pad for further processing.

Load Data Module

Figure 7 load processed data from excel interface.

Figure 7 shows the interface for loading excel data set into the system for support vector machine analysis. To prepare the data, it has to be classified and prepared for analysis. To do this, the study had to use the various

affective states as part of the classification object. An XML file is prepared with the following classification pointers:

- Anger => 1
- Fear => 2
- Disgust => 3
- Happiness => 4 and
- Sadness => 5

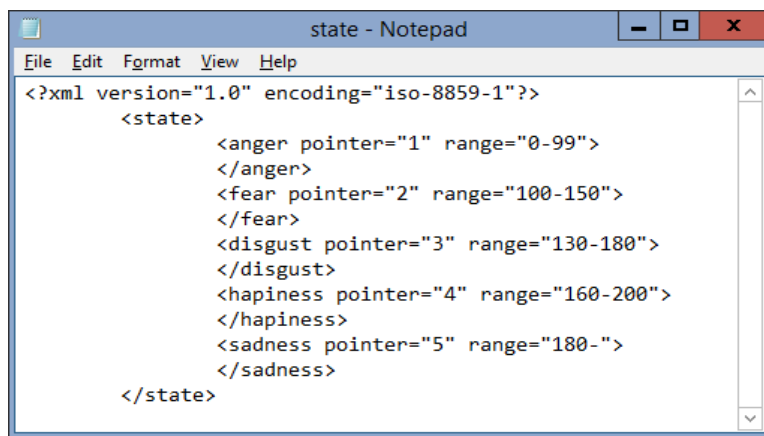


Figure 8 XML file for data classification.

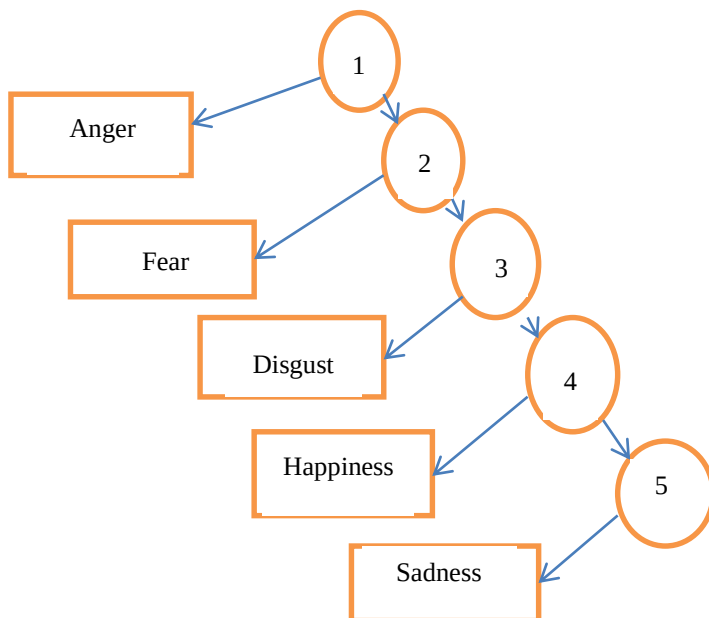


Figure 9 sample XML classification decision tree.

The captured keystrokes are then transferred to an excel file for uploading into a database for analysis. The xml file forms part of the training data for the support vector machine. The support vector machine is then used to analyze the data after it has been trained with the classifier.

Table 1 extracted keystrokes

The logged latencies which were captured using the keystrokes capture module are processed by extracting the keystroke features that are relevant to the research work. Table 1 shows the features of the keystrokes that were extracted from the log file of the keystroke capture application. The main ideas here is to capture feature from the latency of the key down and the key up events of the user of the application. Hence, the study want to know how long the user dwelt on a particular key or group of key in a given time (min_dwell and max_dwell). The study also want to know how fast is the movement of the user from one key to the next key or group of keys in a given time interval (min_flight and max_flight) and the mode of the dwell and flight from key-to-key and also the range of a key down event of one key to the key down event of another key. These features, along with the classification of affective states used in the XML file are used to construct a decision tree that is used by the support vector machine algorithm to analyze the data set.

The Support Vector Machine Module

This is the analysis module. To perform the support vector machine analysis, the data has to be loaded from the database and the SVM algorithm applied on the data. The SVM was trained by the researcher an given a set of

training example which comprise of the classifier and some data, which each of the data marked as belonging to one or the other of five categories of affective states, the SVM algorithm then builds a model that assigns new examples to one of the five categories specified in the classifier,

of data uploaded. Figure 11 – 16 shows the chart for each affective states.

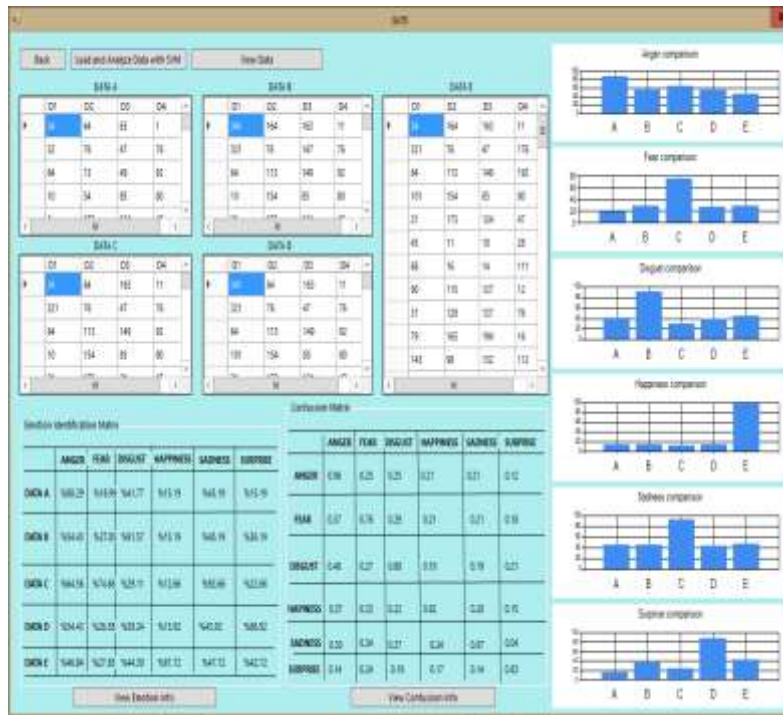


Figure 10 the analysis interface after SVM analysis of the data

Hence, the SVM model is a representation of the examples as points in space, mapped so that the examples of the separate categories are divided by a clear gap that is as wide as possible. New examples are then mapped into that same space and predicted to belong to a category based on which side of the gap they fall. Figure 10 shows the analysis interface after the SVM analysis of the supplied data set.

The Chart Module

This section shows the chart module which shows the chart of comparing different affective state for the classes

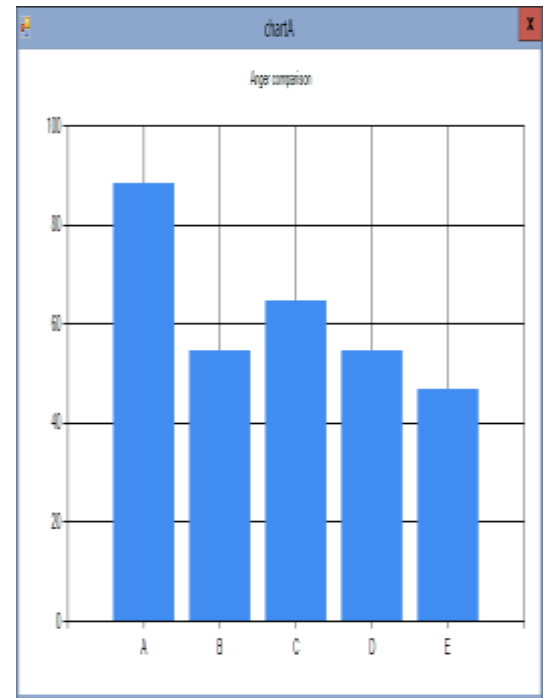


Figure 11 emotion identification comparison for anger

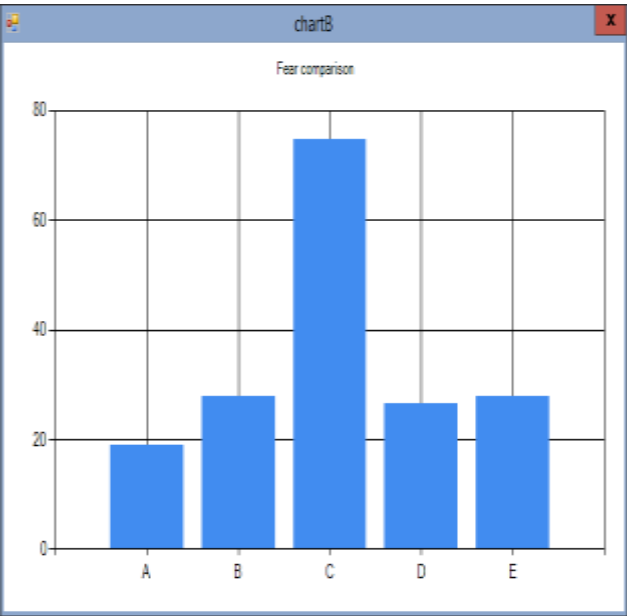


Figure 12 emotion identification comparison for fear

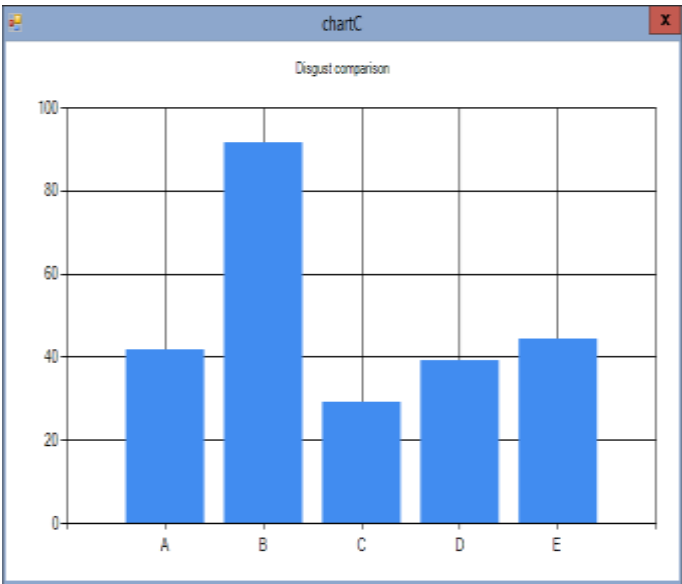


Figure 13 emotion identification comparison for disgust

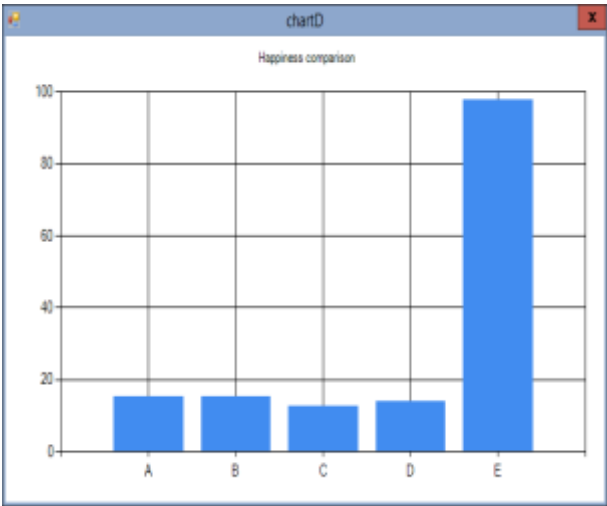


Figure 14 emotion identification comparison for happiness

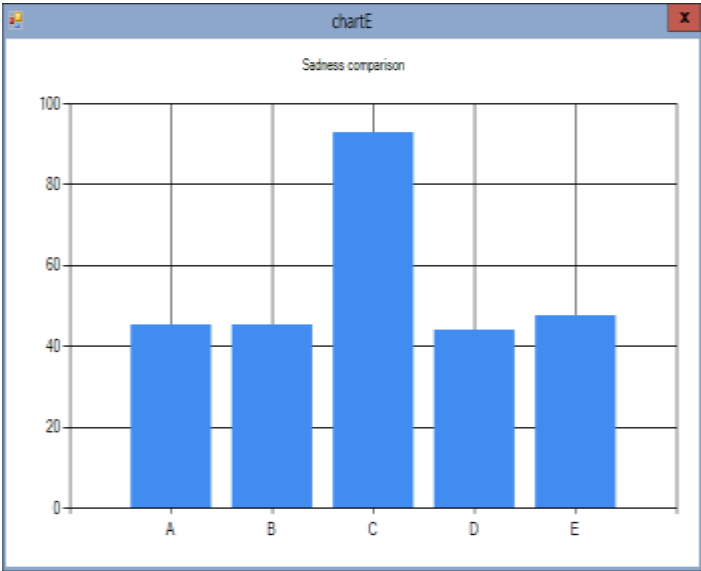


Figure 16 emotion identification comparison for surprise

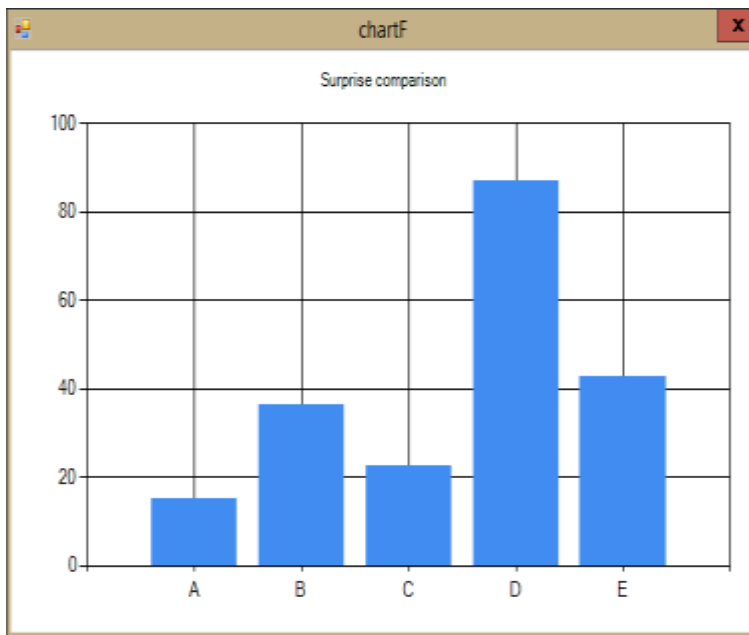


Figure 17 emotion identification matrix

C.THE RESULT

Emotion Identification Matrix

This section of the work discusses the result of the research work. Figure 18 shows the emotion identification matrix from the support vector machine algorithm analysis. The table shows a 5 x 5 matrix with the data sets as the rows and the affective states as the column.

For Data set A

It can be seen from the matrix that the data for data set A has 88.29% of anger identified from the SVM analysis. It can also be seen that happiness seems to be very low with 15.19% which indicates that a person cannot be both angry and happy at the same time. It can also be observed that “disgust” and “sadness” both have 41.77% and 45.19% identification which may suggest a correlation between anger and both affective states.

For Data set B

It can be seen from the emotion identification matrix that the data for data set B after the SVM analysis has 91.57% of disgust identified. It can also be observed that both “anger” and “sadness” both with 54.53% and 43.92% identification also further suggest a correlation between the three (3) affective states.

For Data set C

It can be seen from the emotion identification matrix that the data for data set C after the SVM analysis has 92.66% of sadness identified. It can also be observed that both “anger” and “fear” both with 64.54% and 74.68% identification shows a strong combination of

anger and fear with particular emphasis on that between “sadness” and “anger” which was also identified in the previous data sets.

For Data set D

It can be seen from the emotion identification matrix that the data for data set D after the SVM analysis has 54.43% of anger identified. It can also be observed that both “disgust” and “sadness” both with 39.24% and 43.92% identification shows a strong combination of anger, disgust and sadness which was also identified in the previous data sets.

For Data set E

It can be seen from the emotion identification matrix that the data for data set E after the SVM analysis has 97.72% of happiness identified. It can also be observed that “fear” with an emotional identification of 27.85% seems to be very low.

Arousal-Valence model of Emotion

The arousal-valence model was adopted in determining the degree of the discovered emotion. Figure 18 shows the chart for the arousal-valence model. From the model, it can be seen that there is a strong correlation between fear

and anger, unhappiness (sadness) and disgust. It can also be seen that anger, fear and disgust are in the same plane. It also shows that while fear has negative valence, anger has positive arousal and hence fear can gradually degenerate to anger. This arousal-valence chart further gives credence to the result gotten from the research work. From the result of data A, it can be seen that the degree of anger identified is 88.29% which is both negative valence and high arousal while sadness (unhappy) and disgust which are both 41.77% and 45.19% are both in the negative valence plane.

The same method of identification was applied to the remaining data set as used by the support vector machine algorithm.

Confusion Matrix

The recognition of the affective states was done with the SVM algorithm but before the application of the algorithm, some preparations were done on the collected data to remove mistakes by taking into account the amount of backspaces that was typed by a user of the designed application. To achieve this, the applications ignore the logging of the backspace and delete keys. There are many different methods to correct these mistakes, but it was not possible to catch all of the possible correction scenarios. Outliers for all of the features that involved multiple keys were calculated to remove these pauses (e.g. digraph latency). They were removed by considering the mean and standard deviation for all keystroke dynamic features. Figure 26 shows the confusion matrix which further justifies the research work, the study tried to compare the identified emotion with the selected emotion by the user of the application to

generate the confusion matrix. It shows the level of accuracy of the applied support vector machine.

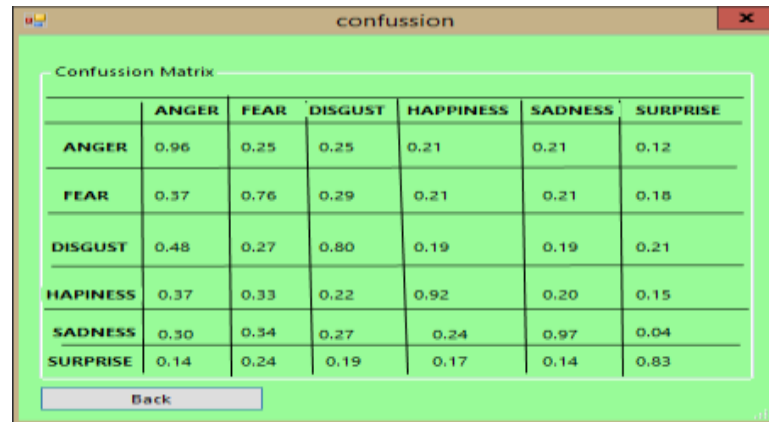


Figure 18: The confusion matrix

Table 2 shows the comparison of results. The research work is compared with [1] and [6]. Figure 19 shows the chart of the comparison. It can be seen that the proposed methodology performs favorably well with existing methodologies especially with [1] work. This is particularly visible in the identification of Anger, Happiness and Sadness.

Table 2 result comparison

EMOTION	Kaveh et al	Clayton et al	Proposed
Anger	93.20%	85.12%	96.10%
Fear/Fright	78.74%	73.07%	76.35%
Happiness	83.50%	58.10%	92.74%
Sadness	91.20%	84.49%	97.18%
Disgust	85.20%	83.35%	80.57%
Surprise	87.34%	82.13%	83.87%

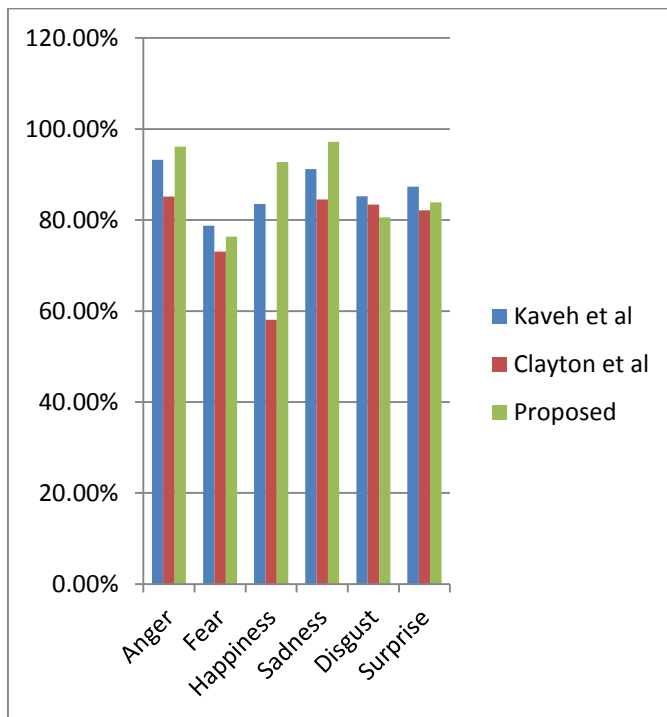


Figure 19 result comparison chart

5. CONCLUSION

First, the arousal-valence stance emotion model was used providing a more discriminative way of emotion definition. This emotion model also allows for regression techniques to be applied to the emotion recognition task. Second, opposed to many other studies that use fixed text, free text data was used in this research. The potential for real-world application is much bigger when free text is used. Third, both general models and individual models were built to investigate the possible advantages of using more user-specific models to make accurate predictions. Fourth, a methodology was presented to use support vector machine for keystroke classification. Finally, an easy-to-use to understand reporting model was used to present the result of the

research work. One of the main problems in this research area is that most studies use a different dataset, different metrics, different features and/or different emotion

models. This causes a lot of difficulties when comparing findings and results. By allowing the study dataset to be used in other studies, the problem of the usage of different datasets and emotion models is solved. One of the limitations of this study was the reliance on the participants for establishing the ground truth. To overcome this, other techniques may be used for identifying the participant's emotional state. Also the usage of textual content can be improved by detecting emotionally charged words, text normalization, stemming techniques and language detection. Finally, clustered models could also be built. This is based on the assumption that every person is unique. However, it is assumed that it is possible that there exist a finite number of clusters or user profiles that comprise people that show similar computer behavior for each emotional state. It is then possible to build one model for each profile. This study presents classification models that are suitable for real-world application and tries to make the data gathering process as real and void of unnecessary stress or triggers that could affect emotional readings.

REFERENCES

1. Bakhtiyari K., Taghavi M, and Husain H (2014) *Implementation of emotional-aware computer systems using typical input devices*. [Intelligent Information and Database Systems: 6th Asian Conference, ACIIDS 2014, Bangkok, Thailand, April 7-9, 2014, Proceedings, Part I](#)(pp.364-374)
2. Bakhtiyari K. and Husain H. (2014) *Fuzzy model in human emotions recognition*. arXiv preprint arXiv:1407.1474
3. Bakhtiyari K and Husain H. (2014) *Fuzzy model of dominance emotions in affective computing*. Neural

- Computing and Applications, vol. 25, no. 6, pp. 1467–1477
4. Cannon W. (1927) *The James-Lange Theory of Emotions: A Critical Examination and an Alternative Theory*. The American Journal of Psychology. 39, 1 (1927), 106-124
 5. Clore G., Ortony A. and Foss M. (1987) *The psychological foundations of the affective lexicon*. Journal of personality and social psychology, vol. 53, no. 4, p. 751
 6. Epp C., Lippold M., and Mandryk R. (2011) *Identifying emotional states using keystroke dynamics*. Conference: Proceedings of the International Conference on Human Factors in Computing Systems, CHI 2011, Vancouver, BC, Canada
 7. Esuli and Sebastiani F. (2006) *Sentiwordnet: A publicly available lexical resource for opinion mining*. Proceedings of LREC, vol. 6. Citeseer, 2006, pp. 417–422.
 8. Forgas J. (1995) *Mood and judgement: The affect infusion model (AIM)*. Psychological Bulletin. 117, 1 (1995), 39-66.
 9. Hektner J., Schmidt J. and Csikszentmihalyi M. (2007) *Experience sampling method: Measuring the quality of everyday life*. Sage
 10. Hernandez-Aguila, M. Garcia-Valdez, and A. Mancilla (2014) *Affective states in software programming: Classification of individuals based on their keystroke and mouse dynamics*. Intelligent Learning Environments, p. 27
 11. Hoffmann H., Scheck A., Schuster T., Walter S., Limbrecht K., Traue H., and Kessler H. (2012) *Mapping discrete emotions into the dimensional space: An empirical approach*. Proceedings of IEEE International Conference on Systems, Man, and Cybernetics (COEX)
 12. Kaernbach C. (2011) *On dimensions in emotion psychology*. Automatic Face & Gesture Recognition and Workshops (FG 2011), IEEE International Conference, pp. 792–796.
 13. Kolakowska A., Landowska A., Szwoch M., Szwoch W and Wróbel M (2014) *Emotion recognition and its applications in Human-Computer Systems Interaction*. [Human-Computer Systems Interaction: Backgrounds and Applications 3](#), pp.51-62
 14. Lee H., Choi Y., Lee S. and Park I. (2012) *Towards unobtrusive emotion recognition for affective social communication*. Consumer Communications and Networking Conference (CCNC), 2012 IEEE. IEEE, 2012, pp. 260–264.
 15. Lee P., Tsui W. and Hsiao T. (2014) *The influence of emotion on keyboard typing: an experimental study using visual stimuli*. Biomedical engineering online, vol. 13, no. 1, p. 81
 16. LiKamWa R., Liu Y., Lane N. and Zhong L. (2011) *Can your smartphone infer your mood*. PhoneSense workshop, pp. 1–5.
 17. LiKamWa R., Liu Y., Lane N. and Zhong L. (2013) *Moodscope: Building a mood sensor from smartphone usage patterns*. Proceeding of the 11th annual international conference on Mobile systems, applications, and services, ACM, 2013, pp. 389–402.
 18. Martin M. (1990) *On the induction of mood*. Clinical Psychology Review, vol. 10, no. 6, pp. 669–697
 19. Milanova M. and Sirakov, N. (2008) *Recognition of Emotional states in Natural Human-Computer Interaction*. IEEE International Symposium on Signal Processing and Information Technology (ISSPIT). IEEE, Ho Chi Minh City
 20. Nahin N., Alam J., Mahmud H. and Hasan K. (2014) *Identifying emotion by keystroke dynamics and text pattern analysis*. Behaviour & Information Technology, vol. 33, no. 9, pp. 987–996
 21. Picard R. (2007). *Affective Computing*. MIT Press.
 22. Schachter S. (1964) *The interaction of cognitive and physiological determinants of emotional state*.

- Advances in Experimental Psychology. 1, (1964), 49-80.
23. Shukla P. and Solanki R. (2013) *Web based keystroke dynamics application for identifying emotional state*. International journal of advanced research in computer science and communication engineering, vol. 2, no. 11, pp. 4489–4493
 24. Strapparava A. Valitutti et al. (2004) Wordnet affect: an affective extension of wordnet. LREC, vol. 4, pp. 1083–1086.
 25. Tsoulouhas G., Georgiou D and Karakos K (2011) *Detection of learner's affective state based on mouse movements*. J. Comput, vol. 3, pp. 9–18
 26. Tsui W., Lee P. and Hsiao T. (2013) *The effect of emotion on keystroke: an experimental study using facial feedback hypothesis*. Engineering in Medicine and Biology Society (EMBC), 2013 35th Annual International Conference of the IEEE. IEEE, 2013, pp. 2870– 2873.
 27. Villani M., Tappert C. et al. (2006) *Keystroke Biometric Recognition Studies on Long-Text Input under Ideal and Application-Oriented Conditions*. Proceedings of the 2006 Conference on Computer Vision and Pattern Recognition Workshop (2006), 39.
 28. Vizer L.M. (2009) *Detecting cognitive and physical stress through typing behavior*. Proceedings of the 27th international conference extended abstracts on Human factors in computing systems (Boston, MA, USA, 2009), 3113-3116.
 29. Vizer, L.M., Zhou, L. et al. (2009) *Automated stress detection using keystroke and linguistic features: An exploratory study*. Int. J. Hum.-Comput. Stud. 67, 10 (2009), 870-886.
 30. Westermann, R., Spies, K. et al. (1996) *Relative effectiveness and validity of mood induction procedures: a meta-analysis*. European Journal of Social Psychology. 26, 4 (1996), 557-580.
 31. Witten I. and Frank E. (2005). *Data Mining: Practical machine learning tools and techniques*. Morgan Kaufmann.
 32. Zimmermann, P., Gomez, P. et al. (2006). *Extending usability: putting affect into the user experience*. Proceedings of NordiCHI'06 (New York, 2006), 27-32.
 33. Zhou L., Burgoon J., Nunamaker J. and Twitchell D. (2004) *Automating linguistics-based cues for detecting deception in text-based asynchronous computer-mediated communications*. Group decision and negotiation, vol. 13, no. 1, pp. 81–106, 2004.
 34. Zimmermann P., Guttormsen S., Danuser B. and Gomez P. (2003) *Affective computing – a rationale for measuring mood with mouse and keyboard*. International journal of occupational safety and ergonomics, vol. 9, no. 4, pp. 539–551